

# Corporate Finance

## Lecture 1: Capital Markets and the Pricing of Risk

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# Plan

- ▶ Learn some general concepts
- ▶ Learn how to compute returns
- ▶ Understand how we measure risk
- ▶ Understand what is an index

# Key concepts

- ▶ Compute/estimate returns and risk
- ▶ Risk-return relationship of assets

# Outline

- A. Risk and return: a first glimpse (Chap. 10.1)
- B. Common return measures (Chap. 10.2)
- C. Historical returns of stocks and bonds (Chap. 10.3)
- D. Common measures of risk (Chap. 10.2)
- E. The historical risk-return trade-off (Chap. 10.4)
- F. Risk and returns of individual stocks (Chap. 10.4)

**But first, let's battle a monster.**



*This is a monster you will find along the course several times*

Initially it may look scary...



But hopefully, you will see that the monster is not that scary



# The monster only means “add these up”

- ▶ The symbol  $\sum$  is the Greek letter S, and S stands for **sum**. It is a shortcut for writing a long addition.
- ▶ Instead of writing the addition in full, we can write it with the symbol:

$$1 + 2 + 3 + 4 + 5 \quad \text{is written} \quad \sum_{i=1}^5 i$$

- ▶ Both mean exactly the same thing: **15**. The symbol saves ink, nothing more.

# How to read the monster

- ▶ Three things are attached to it:
  - ▶ Below: where to **start** counting, here  $i = 1$ .
  - ▶ Above: where to **stop**, here 5.
  - ▶ To the right: **what to add**, here just  $i$  itself.
- ▶ So you take the recipe on the right, put in  $i = 1$ , then  $i = 2$ , and so on until you reach the number on top, and add up everything you wrote:

$$\sum_{i=1}^5 i = \underbrace{1}_{i=1} + \underbrace{2}_{i=2} + \underbrace{3}_{i=3} + \underbrace{4}_{i=4} + \underbrace{5}_{i=5} = 15$$

## The recipe can be anything

- ▶ The letter  $i$  is just a counter. Whatever sits to the right of the symbol is the recipe, and the counter is replaced by 1, then 2, then 3...
- ▶ Adding the squares of the first four numbers:

$$\sum_{i=1}^4 i^2 = 1^2 + 2^2 + 3^2 + 4^2 = 1 + 4 + 9 + 16 = 30$$

- ▶ In finance the recipe is usually a cash flow to be discounted:

$$\sum_{i=1}^3 \frac{C_i}{(1+r)^i} = \frac{C_1}{1+r} + \frac{C_2}{(1+r)^2} + \frac{C_3}{(1+r)^3}$$

## Summing up to $n$

- Often we do not know how many terms there are, so we call that number  $n$ :

$$\sum_{i=1}^n \frac{C_i}{(1+r)^i} = \frac{C_1}{1+r} + \frac{C_2}{(1+r)^2} + \dots + \frac{C_n}{(1+r)^n}$$

- It says: start at the first year, stop at year  $n$ , and add up every discounted cash flow in between.
- A project paying for 5 years has  $n = 5$ ; one paying for 30 years has  $n = 30$ . The formula does not change, only where we stop.

## Summing up to infinity

- Sometimes the payments never stop, for instance a company that is expected to pay a dividend every year forever. Then we write  $\infty$  on top:

$$\sum_{i=1}^{\infty} \frac{C}{(1+r)^i} = \frac{C}{1+r} + \frac{C}{(1+r)^2} + \frac{C}{(1+r)^3} + \dots$$

- Adding infinitely many numbers does not have to give an infinite total. Each term here is smaller than the one before, because we divide by  $(1+r)$  once more every year.
- The terms shrink fast enough that the total settles on a finite number. This is the perpetuity, and it is why a share paying dividends forever still has a finite price.

# Why the total can be finite

- ▶ A simpler example of the same idea: keep adding halves of what is left:

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots$$

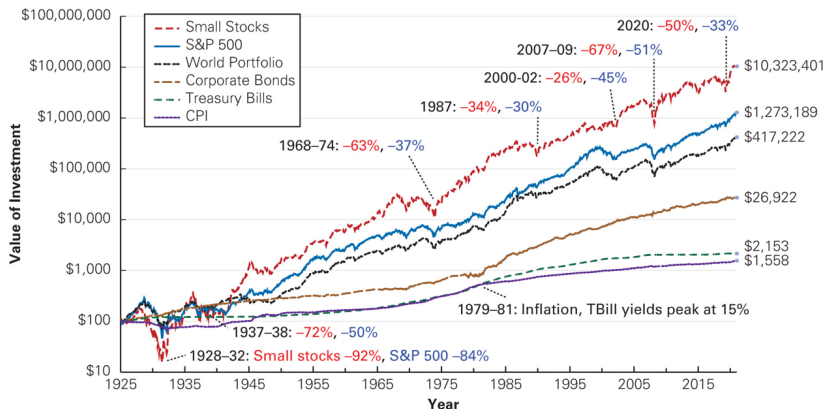
- ▶ After two terms you are at 0.75, after three at 0.875, after four at 0.9375. The total never passes 1, it just gets closer and closer.
- ▶ The monster is tamed: an endless sum of shrinking terms adds up to a number you can actually put in a valuation.

## A. Risk and return: a first glimpse

- ▶ Standard & Poor's 500: 90 U.S. stocks up to 1957 and 500 after that. Leaders in their industries and among the largest firms traded on U.S. Markets.
- ▶ Small stocks: securities traded on the NYSE with market capitalizations in the bottom 20%.
- ▶ MSCI World Index (World portfolio)
- ▶ Corporate Bonds: long-term, AAA-rated U.S. corporate bonds with maturities of approximately 20 years.
- ▶ Treasury Bills: an investment in three-month Treasury bills.
- ▶ CPI (consumer price index)

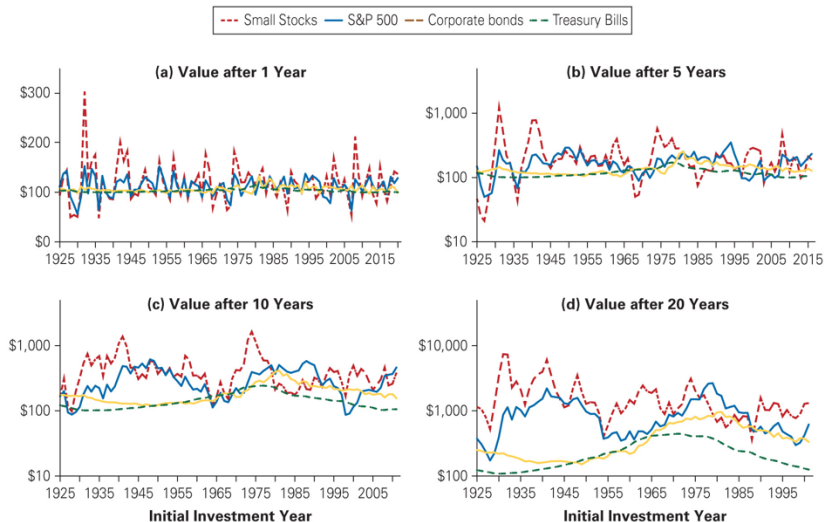
⇒ We ignore transaction costs and assume that interest and dividends are reinvested.

# Value of \$100 invested at the end of 1925



Source: CRSP, Standard and Poor's, MSCI, Global Financial Data.

# Value of \$100 invested in different years and for differing horizons



Source: CRSP, Standard and Poor's, MSCI, Global Financial Data.

## B. Common return measures

How to determine the return of a financial asset?

- ▶ Returns from dividends and capital gains
- ▶ Calculations with reinvested dividends
- ▶ Stock market indices
- ▶ Histograms of historical returns
- ▶ Historical volatility
- ▶ Arithmetic mean versus geometric mean

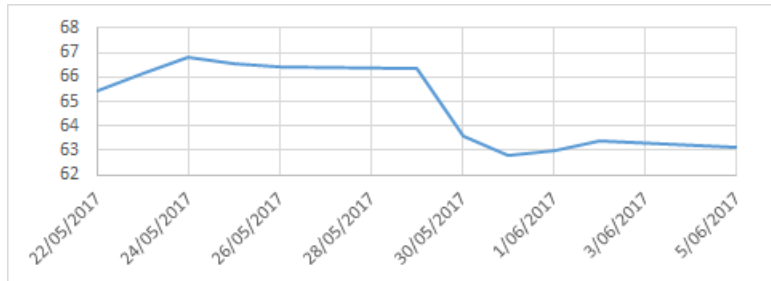
# Dividend return and capital gains

- ▶ The realized return of a stock between two dates consists of:
  - ▶ capital gains and
  - ▶ dividends

$$\begin{aligned} R_{t+1} &= \frac{\text{Wealth at period end}}{\text{Initial investment}} - 1 \\ &= \frac{P_{t+1} + Div_{t+1} - P_t}{P_t} \\ &= \underbrace{\frac{P_{t+1} - P_t}{P_t}}_{\text{Capital Gain Rate}} + \underbrace{\frac{Div_{t+1}}{P_t}}_{\text{Dividend Yield}} \end{aligned}$$

# Stock prices and dividend payment: the example of BNP Paribas

A dividend of EUR 2.7 is detached from the share price on May 30th, 2017



Source: Yahoo Finance

## How to calculate the average return on an investment?

- What is the better investment (Investment at the end of year N until year N+2)?

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|              |     |      |       |         |
|--------------|-----|------|-------|---------|
| Investment 1 |     |      |       |         |
| Value        | 100 | 50   | 100   | Average |
| Return       |     | -50% | +100% | +25 %   |
| Investment 2 |     |      |       |         |
| Value        | 100 | 110  | 121   |         |
| Return       |     | +10% | +10%  | +10%    |

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- The arithmetic average is misleading ...

## Geometric mean (annualized), or CAGR

- ▶ The geometric mean is calculated as

$$\text{Geometric mean} = \left( \frac{\text{Wealth at the end of n years}}{\text{Initial investment}} \right)^{\frac{1}{n}} - 1$$

- ▶ or Geometric mean =  $((1 + R_1)(1 + R_2) \dots (1 + R_n))^{\frac{1}{n}} - 1$

## To sum up: calculation of returns

- ▶ Depending on the situation, it is best to calculate mean returns differently. In general:
  - ▶ Intertemporal average or realized returns: geometric mean (=IRR of the investment, or “compound annual return”)
  - ▶ Mean return of multiple investments or expected return: arithmetic mean
- ▶ The two means are equal if annual returns are constant. Otherwise the arithmetic mean is always larger than the geometric mean.
- ▶ The geometric mean is approximately equal to the arithmetic mean -  $\frac{1}{2}$  variance.

## Compounding returns

- ▶ The returns over multiple periods need to be compounded to obtain the return over a longer horizon.
- ▶ Example: Over 4 consecutive periods an investment generates the returns: 6.1%, -5.32%, 6.72% and 8.93%. The overall return is:

$$\begin{aligned}R_{1 \rightarrow 4} &= (1 + R_1) (1 + R_2) (1 + R_3) (1 + R_4) - 1 \\&= (1 + 0.061) (1 - 0.0532) (1 + 0.0672) (1 + 0.0893) - 1 \\&= 16.7\%\end{aligned}$$

- ▶ Remarks:
  - ▶ You must not take the sum of the returns!
  - ▶ The length of the periods does not matter here.

## Example: accounting for reinvested dividends to compute the returns on Total

What was the total return of the Total stock in 2016 (rounded to 4 decimal points)?

| Date       | Price<br>(in EUR) | Dividend<br>(in EUR) | Return |
|------------|-------------------|----------------------|--------|
| 01/01/2016 | 40.89             |                      |        |
| 21/03/2016 | 41.33             | 0.61                 | 2.57%  |
| 06/06/2016 | 42.47             | 0.61                 | 4.23%  |
| 27/09/2016 | 40.77             | 0.61                 | -2.57% |
| 21/12/2016 | 47.60             | 0.61                 | 18.25% |
| 31/12/2016 | 48.72             |                      | 2.35%  |

## Returns and “stock splits”

- ▶ Companies sometimes decide to do “stock splits”, or “reverse stock splits”, which need to be taken into account when returns are calculated.
- ▶ Example:
- ▶ You bought 100 stocks of CIGNA Corp. on August 5th, 2006 for a price of \$155.59. One year later, the stock is worth \$50.18, but the company has done a “3:1 stock split” so you now own 300 stocks. Over this period, you received a total amount of dividends of \$8.5 (on all your stocks). Your total return is

$$\frac{300 \times 50.18 + 8.5 - 100 \times 155.59}{100 \times 155.59} = -3.2\%.$$

# Value of a stock and performance index

- ▶ The previous figures don't correspond to the evolution of a specific stock price.
  - ▶ They represent the evolution of an investment in the stock.
- ▶ They are a “performance index” and/or use “adjusted prices”.
- ▶ The adjusted stock price accounts for dividends and stock splits. They are calculated such that
  - ▶ The last value corresponds to the stock price.
  - ▶ The previous values mirror the evolution of an investment in the stock.

## Example: the long-term return on a stock

- ▶ On January 2nd, 1970, the Coca Cola stock price was worth \$82.25. On June 30th, 2017, the stock price was worth \$44.85. Good or bad investment?
- ▶ Coca Cola has done 5 stock splits and made many dividend payments, the “adjusted close” on 2.1.1970 was 0.85.
- ▶ Hence, the return on the overall period 1970-2017 is  $44.85/0.85 - 1 = 5176\%$ .
- ▶ The geometric annual mean is  $(44.85/0.85)^{1/47} - 1 = 8.80\%$ .

# Stock price indices

- ▶ The stock price indices aggregate the information about the evolution of a stock exchange by depicting the evolution of a portfolio of stocks.
- ▶ Example: CAC 40, DAX, S&P 500, FTSE100 (Footsie), Nasdaq 100, Dow Jones Industrial average, Hang Seng, Nikkei 225 etc.
- ▶ Some of these indices are very old!
  - ▶ Dow Jones has published its index since 1896 in New-York
- ▶ **Attention:** These indices can be calculated as “total return index” including dividends or “price index” without dividends! Some indices exist in different versions.

# CAC40 and CAC40 GR (gross return)



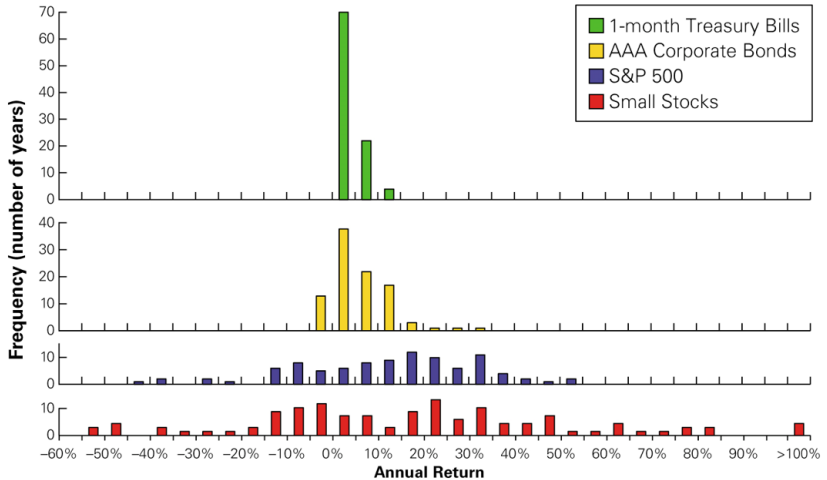
## C. Historical returns of stocks and bonds

Annual mean return

$$\overline{R} = \frac{1}{T} \sum_{t=1}^T R_t = \frac{1}{T} (R_1 + \dots + R_T)$$

| Investment      | Average Annual Return |
|-----------------|-----------------------|
| Small stocks    | 18.6%                 |
| S&P 500         | 12.2%                 |
| Corporate bonds | 6.2%                  |
| Treasury bills  | 3.3%                  |

# Return histograms of different US portfolios (1926-2021)



## D. Common measures of risk

- ▶ There is no perfect and generally accepted risk measure.
- ▶ In finance, risk is measured by volatility.
- ▶ Volatility = standard deviation of the returns

First calculate the variance, then take its square root to get the volatility:

$$\text{Var}[R] = \frac{1}{T-1} \sum_{t=1}^T (R_t - \bar{R})^2 \qquad \sigma_R = \sqrt{\text{Var}[R]}$$

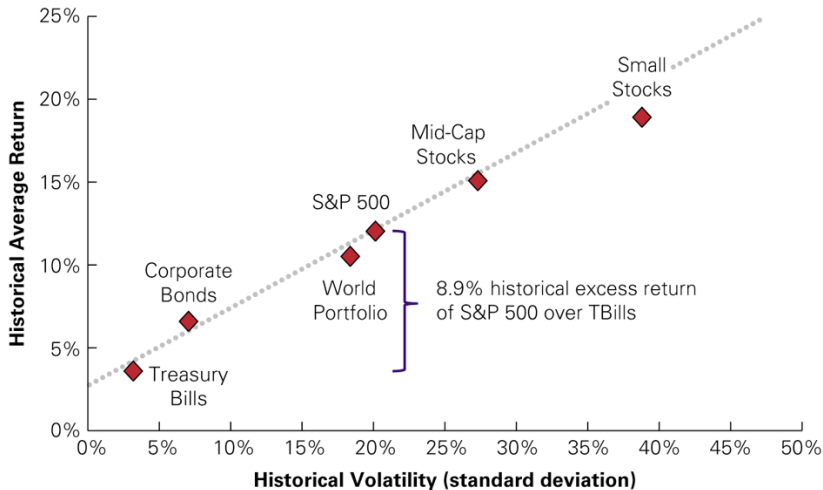
- ▶ Attention:
  - ▶ The volatility is a symmetric risk measure. It takes into account the risk to win as well as the risk to lose!
  - ▶ Other measures capture only the risk of losses (downside risk), e.g. VaR (“value at risk”).

## E. The historical risk and return trade-off

| Investment                 | Return Volatility<br>(Standard Deviation) | Excess Return (Average Return<br>in Excess of Treasury Bills) |
|----------------------------|-------------------------------------------|---------------------------------------------------------------|
| Small stocks               | 38.6%                                     | 15.3%                                                         |
| S&P 500                    | 19.7%                                     | 8.9%                                                          |
| Corporate<br>bonds         | 6.3%                                      | 2.9%                                                          |
| Treasury bills<br>(30-day) | 3.1%                                      | ---                                                           |

Excess return = return in excess of the risk-free rate  
= risk premium

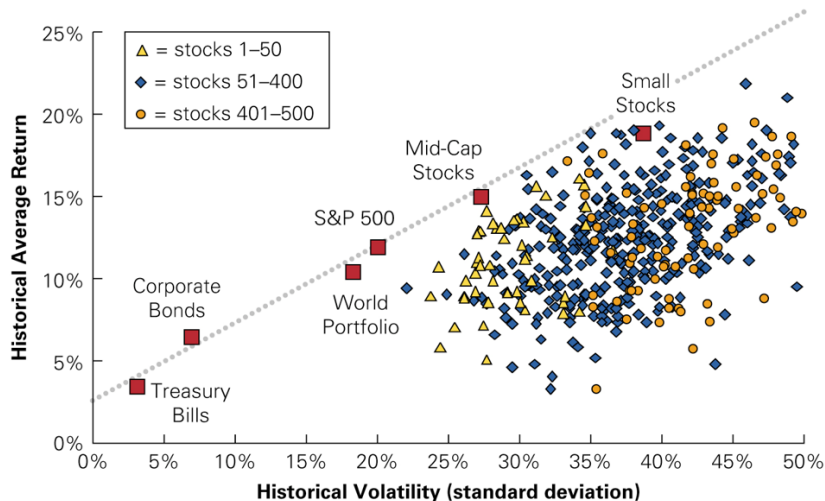
# Return/risk of different US portfolios



Source: CRSP, Morgan Stanley Capital International

## F. Risk and returns of individual stocks

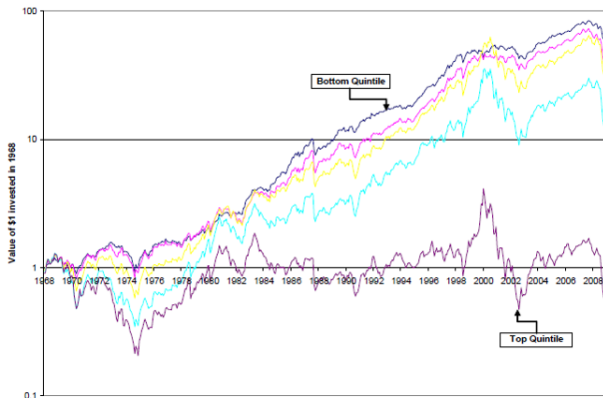
Volatility and realized returns on S&P 500 stocks



Source: CRSP, Morgan Stanley Capital International

# What's "new"? The low-risk anomaly

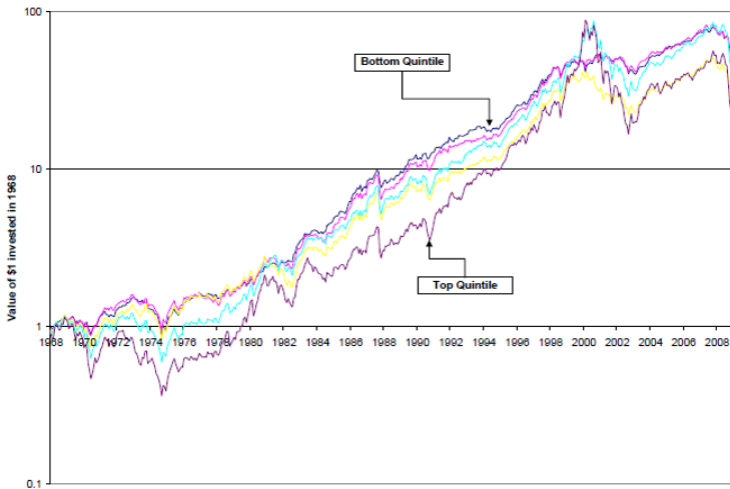
All stocks:



Baker, Bradley and Wurgler (2011).

# What's "new"? The low-risk anomaly (con't)

Top 1000 stocks by MktCap



Baker, Bradley and Wurgler (2011).

# Quiz

- ▶ A stock is worth \$1 at the beginning of the year. At the end of the year and after a reverse stock split of 1:10 and a dividend of \$0.03 (before the split), the stock is worth \$11. What is the return on the investment?
- ▶ The value of a stock is halved over six month (without stock split and dividends).
  - ▶ What is the return on the investment over the six months?
  - ▶ What is the annualized return?
- ▶ The value of a stock loses 20% and then gains 20% (without stock split and dividends).
  - ▶ What is the average arithmetic return?
  - ▶ What is the average geometric return?

# Quiz

- ▶ Your banker explains to you that stocks always offer a better return than your investment in treasury bills.
  - ▶ Is this true?
  - ▶ Is it a reason to invest in stocks?
- ▶ Can volatility become negative?
- ▶ Is the return on a stock with more risk always higher than the return on a stock with less risk?

## Critical thinking

- ▶ A bond is worth \$97 (for a nominal value of \$100) at the beginning of the year. The bond pays an annual coupon of 4.5%, and is worth \$102 at the end of the year. What is the return on the investment?
- ▶ A company's stock price increases by 10% over the year, but the return on equity (ROE) of the company is -30%. Is there a contradiction?